



ENSAE - 3rd year, Master 2
Exam 2024-2025

RISK MEASURES

Closed book examination, no calculator - 2 hours

The exercises are independent. Each question counts for approximately 2 points. You can answer in English or in French.

Exercise 1 (General questions about the course)

1. Compare and contrast Value-at-Risk (VaR) with Expected Shortfall (ES). Is ES always greater than VaR? Is ES always "better" than VaR?
2. Compute the cumulative distribution function F and the quantile function F^- of a random variable that takes the value -1 with probability 1/3, is uniformly distributed between 0 and 1 with probability 1/3 and takes the value 2 with probability 1/3.
3. In standard notation, is it possible to have $F(x) < \alpha$ and $x \geq F^-(\alpha)$? Is it possible to have $F(F^-(\alpha)) < \alpha$? to have $F(F^-(\alpha)) > \alpha$?
4. Explain how the convexity or concavity of G , the distortion function, reflects an optimistic or pessimistic attitude toward risk in the DRM

$$r(F, G) = \int_0^1 F^{-1}(1-u) dG(u).$$

Give an example of distortion function G .

5. Explain all the notations and assumptions needed for the following result

$$\sqrt{n}(\xi_{n,\alpha} - \xi_\alpha) \xrightarrow{\mathcal{L}} \mathcal{N}\left(0, \frac{\alpha(1-\alpha)}{f^2(\xi_\alpha)}\right).$$

Interpret the variance.

6. Let $X_1, \dots, X_n$ be an iid sequence of continuous real variables with a spherical distribution. What does this mean? What is the relationship between the density of $|X_i|$ and that of X_i ? For $\alpha \in (0, 1/2)$, what is the relationship between the $1 - 2\alpha$ quantile of $|X_i|$ and the α quantile ξ_α of X_i ? Use the previous question to suggest an estimator of ξ_α that is more accurate than the empirical quantile of the X_i 's.

Exercise 2 (Dynamic Systemic Risk Measures)

Let a bivariate real process $\epsilon_t = (\epsilon_{1t}, \epsilon_{2t})'$ of returns, and an information set $\mathcal{F}_{t-1}$ containing $\{\epsilon_u, u < t\}$. For $i = 1, 2$, the dynamic VaR of $\epsilon_i = (\epsilon_{it})_{t \in \mathbb{Z}}$ at risk level $\alpha \in (0, 1)$ is defined as the opposite of the α -quantile of the conditional distribution of ϵ_i :

$$\text{VaR}_t^{\epsilon_i}(\alpha) = -\inf\{z_t : P_{t-1}[\epsilon_{it} \leq z_t] \geq \alpha\}$$

where P_{t-1} denotes the historical distribution conditional on $\mathcal{F}_{t-1}$. The dynamic CoVaR of ϵ_1 relative to ϵ_2 at risk levels $\alpha, \alpha' \in (0, 1)$ is defined as

$$\text{CoVaR}_t^{\epsilon_1|\epsilon_2}(\alpha, \alpha') = -\inf\{z_t : P_{t-1}[\epsilon_{1t} \leq z_t | \epsilon_{2t} \leq -\text{VaR}_t^{\epsilon_2}(\alpha')] \geq \alpha\}.$$

1. Suppose ϵ_1 represents the returns of a particular financial institution and ϵ_2 represents the returns of the "system". The event

$$\{\epsilon_{2t} \leq -\text{VaR}_t^{\epsilon_2}(\alpha')\}$$

can then be interpreted as "the system is in distress". How do you interpret a particularly large value of $\text{CoVaR}_t^{\epsilon_1|\epsilon_2}(\alpha, \alpha')$? a large value of $\text{CoVaR}_t^{\epsilon_2|\epsilon_1}(\alpha, \alpha')$? Is there a relationship between $\text{CoVaR}_t^{\epsilon_1|\epsilon_2}(\alpha, \alpha')$ and $\text{CoVaR}_t^{\epsilon_2|\epsilon_1}(\alpha, \alpha')$? How would you measure the systemic risk of an institution?

Let (η_{1t}, η_{2t}) be an iid process with $E\eta_{it} = 0$ and $E\eta_{it}^2 = 1$. Let $\xi_\alpha^{(i)} = \inf\{x : G^{(i)}(x) \geq \alpha\}$ be the α -quantile of the cumulative distribution function (cdf) $G^{(i)}$ of η_{it} . We will call co-cdf the function

$$F(x|y) = P[\eta_{1t} \leq x | \eta_{2t} \leq y]$$

for $x \in \mathbb{R}$, $y \in \mathbb{R}$ such that $P[\eta_{2t} \leq y] \neq 0$. We then call $u(\alpha, \alpha')$ the co-quantile of η_1 and η_2 at levels (α, α') , defined by

$$u(\alpha, \alpha') = \inf\left\{x : F\left(x | \xi_{\alpha'}^{(2)}\right) \geq \alpha\right\}.$$

Table 1: Backtesting results for H_0 (Frequency of Null Rejection over a selection of 426 US Stocks) at the nominal level $\alpha_0 = 5\%$ with the S&P 500 as the conditioning asset

Model	Unconditional Coverage Test	Independence Test
Parametric Gaussian	0.110	0.110
Semi-Parametric FHS	0.066	0.063

2. In this question we assume that (η_{1t}, η_{2t}) is Gaussian with $\rho = E\eta_{1t}\eta_{2t}$. Let Φ and ϕ be the cdf and density of the standard Gaussian distribution. Show that we have

$$F(y | x) = \frac{1}{\Phi(x)} \int_{-\infty}^x \Phi\left(\frac{y - \rho u}{\sqrt{1 - \rho^2}}\right) \phi(u) du.$$

How are $\xi_\alpha^{(i)}$ and $u(\alpha, \alpha')$ defined?

Assume that the ϵ_i 's satisfy location-scale models of the form

$$\epsilon_{it} = \mu_{it} + \sigma_{it}\eta_{it}, \quad t \in \mathbb{Z}, \quad i = 1, 2,$$

where μ_{it} and σ_{it} are $\mathcal{F}_{t-1}$ -measurable and η_{it} are independent of $\mathcal{F}_{t-1}$. Assume the existence of a consistent estimator $(\hat{\mu}_{it}, \hat{\sigma}_{it})$ of (μ_{it}, σ_{it}) (e.g. based on a parametric specification of these conditional moments).

3. Express $\text{VaR}_t^{\epsilon_2}(\alpha')$ and $\text{CoVaR}_t^{\epsilon_1|\epsilon_2}(\alpha, \alpha')$ as function of μ_{it} , σ_{it} and $u(\alpha, \alpha')$. Based on the residuals $\hat{\eta}_{it} = (\epsilon_{it} - \hat{\mu}_{it})/\hat{\sigma}_{it}$, propose an FHS-type estimator of $\text{CoVaR}_t^{\epsilon_1|\epsilon_2}(\alpha, \alpha')$.
4. Table 1 shows results of backtests of the null H_0 of iid violation of probability α for two estimated CoVaRs for series of daily log returns of 426 US stocks over a 13-year period (3,000 in-sample data points vs. 250 out-of-sample observations). What are the two tests in this table? What conclusion can we draw from this table? Which method works better?